

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

3 Probability and Set Theory

3.1 Setup

Definition 3.1.1. An **experiment** is any activity or process whose outcome is subject to uncertainty.

Definition 3.1.2. The **sample space** of an experiment, denoted by $\mathbb{S}$, is the set of all possible outcomes of that experiment.

- A simplest example of an experiment is a coin toss, whose sample space is $\mathbb{S} = \{head, tail\}$ - or in short, $\{H, T\}$.
- In this section, all things are based on a given experiment with sample space denoted as $\mathbb{S}$.

Definition 3.1.3.0. An **event** is any collection (subset) of outcomes contained in the sample space.

- The sample space itself is also an event.

Definition 3.1.3.1. An event is **simple** if it consists of exactly one outcome.

Definition 3.1.3.2. An event is **compound** if it consists of more than one outcome.

- Consider an experiment in which two coins are simultaneously flipped, its sample space is $\mathbb{S} = \{HH, TT, HT\}$, a simple event of this experiment is $\{HH\}$, and a compound event of it is $\{TT, HT\}$.
 - However, the sample space of an experiment where two coins are sequentially flipped is not the same as that above.

Definition 3.1.4. Event relations/operations in the context of set theory:

1. The **complement** of an event A , denoted by A' or A^C , is the set of all outcomes in the sample space $\mathbb{S}$ that are not contained in A .

2. The **union** of two events A and B , denoted by $A \cup B$ and read as “ A or B ”, is the event consisting of all outcomes that are either in A or in B (and thus, “or” in both events) - that is, all outcomes in at least one of the events.
3. The **intersection** of two events A and B , denoted by $A \cap B$ and read as “ A and B ”, is the event consisting of all outcomes that are in both A and B .
4. The **null event**, denoted as $\emptyset$, is the event consisting of no outcome of an experiment.
5. For any $n \in \mathbb{N}_+$ so that $n \geq 2$ and a set of events $\{A_i\}_{i=1}^n$, the events are said to be **mutually exclusive** or **(pairwise) disjoint** if

$$A_i \cap A_j = \emptyset \text{ for any } i, j \in \{1, 2, \dots, n\} \text{ and } i \neq j.$$

- We can use Venn diagrams - a popular method of presenting logical relations among sets - to pictorially represent the above relations among events.

3.2 Axioms and Properties

Definition 3.2.1. For any event $A \subset \mathbb{S}$, the **probability** of event A , denoted as $P(A)$, is the precise measure of the chance that A will occur in the experiment.

- The probability of an event can be understood as the “limiting” or “long-run” relative frequency of the event.¹

Axiom 3.2.2.1. For any event $A (\subset \mathbb{S})$, $P(A) \geq 0$.

Axiom 3.2.2.2. $P(\mathbb{S}) = 1$.

Axiom 3.2.2.3. For any countably infinite collection of disjoint events $\{A_i\}_{i \in \mathbb{N}_+}$,

$$P\left(\bigcup_{i \in \mathbb{N}_+} A_i\right) = \sum_{i \in \mathbb{N}_+} P(A_i).$$

Proposition 3.2.3. $P(\emptyset) = 0$.

- Proof idea: Consider $A_i = \emptyset$ for all $i \in \mathbb{N}_+$.
- Given Proposition 3.2.3, we know that Axiom 3.2.2.3 can be extended to any finite collection of disjoint events - in the case of n disjoint events, consider $A_i = \emptyset$ for all $i > n$.

Proposition 3.2.4. For any event A , $P(A') = 1 - P(A)$.

- Proof idea: With Axiom 3.2.2.3, consider $A_1 = A$, $A_2 = A'$, and $A_i = \emptyset$ for all $i \geq 3$.

Corollary 3.2.5. For any event A , $P(A) \leq 1$.

¹A reference that is rather technical is Law of Large Numbers (LLN) - which is not required in this course.

- Proof idea: Directly by Proposition 3.2.4 and Axiom 3.2.2.1.

Proposition 3.2.6. For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- Proof idea: Partition $A \cup B$ into A and $A' \cap B$, while (1) $A' \cap B$ and $A \cap B$ are disjoint and (2) $(A' \cap B) \cup (A \cap B) = B$.
– Can also be illustrated by Venn diagram.

Corollary 3.2.7. For any three events A , B , and C ,

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

- Proof idea: Partition $A \cup B \cup C$ into $A \cup B$ and $A' \cap B' \cap C$. Then, apply Proposition 3.2.6 to $A \cup B$, and apply the finite version of Axiom 3.2.2.3 three times given (1) C can be partitioned into $A \cap B \cap C$, $A \cap B' \cap C$, $A' \cap B \cap C$, and $A' \cap B' \cap C$, (2) $A \cap C$ can be partitioned into $A \cap B \cap C$ and $A \cap B' \cap C$, and (3) $B \cap C$ can be partitioned into $A \cap B \cap C$ and $A' \cap B \cap C$.
- Corollary 3.2.7 is the three-event extension of Proposition 3.2.6. In fact, Proposition 3.2.6 can be extended to situations with any finite number of events.

3.3 Conditional Probability and Independence

Definition 3.3.1. For any two events A and B with $P(B) > 0$, the **conditional probability** of A given that B has occurred is

$$P(A|B) = \frac{P(A \cap B)}{P(B)},$$

where B is called conditional event.

- Definition 3.3.1 directly implies the **multiplication rule**: For any two events A and B ,

$$P(A \cap B) = P(A|B)P(B).$$

Definition 3.3.2. For a set of events $\{A_i\}_{i=1}^n$, the events are **exhaustive** if

$$\bigcup_{i=1}^n A_i = \mathbb{S}.$$

- In other words, events are exhaustive if the probability that (at least) one of them happens is 1.

Theorem 3.3.3. (Law of Total Probability) For any set of mutually exclusive and exhaustive events $\{A_i\}_{i=1}^n$ and any other event B ,

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i).$$

- Proof idea: Because $\{A_i\}_{i=1}^n$ is a set of mutually exclusive and exhaustive events - i.e., it is a partition of the sample space (and vice versa) - B can be partitioned into $\{B \cap A_i\}_{i=1}^n$. Then, the proof concludes by applying the multiplication rule to each of $B \cap A_i$.

Theorem 3.3.4. (Bayes' Theorem) For any set of mutually exclusive and exhaustive events $\{A_i\}_{i=1}^n$ and any other event B with $P(B) > 0$,

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)} \text{ for each } j \in \{1, 2, \dots, n\}.$$

- Proof idea: Directly by the multiplication rule and Theorem 3.3.3.
- In the context of Bayes' Theorem, the conditional probability $P(A_j|B)$ is usually called the **posterior probability** of A_j given that B has occurred. By contrast, the unconditional probability $P(A_j)$ is usually called the **prior probability** of A_j . Here, prior and posterior is relative to the occurrence of the conditioning event B .

Definition 3.3.5. Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$ - and they are **dependent** if otherwise.

- A and B are independent if $P(A) = 0$ or $P(B) = 0$.
- When $P(A) > 0$ and $P(B) > 0$,
 - A and B are independent if and only if (iff) $P(A|B) = P(A)$ (and, equivalently, $P(B|A) = P(B)$) - that is, the occurrence of one of the events doesn't affect the probability of the other;
 - A and B are dependent if they are mutually exclusive.

Definition 3.3.6. For some $n \in \mathbb{N}_+$ and $n \geq 2$, events $A_1, A_2, \dots, A_n$ are **mutually independent** if for any $k \in \{2, 3, \dots, n\}$ and any subset of indices $\{i_1, i_2, \dots, i_k\}$,

$$P\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k P(A_{i_j}).$$

3.4 Counting Rule

Proposition 3.4.1. For a ordered set of $k \in \mathbb{N}_+$ elements with $n_i \in \mathbb{N}_+$ possible values of the i^{th} element for each $i \in \{1, 2, \dots, k\}$, the number of possible realizations of the ordered set is

$$\prod_{i=1}^k n_i.$$

Definition 3.4.2. A **permutation** is an ordered set. The number of permutations of size $k \in \mathbb{N}_+$ that can be formed from a set of size $n \in \mathbb{N}_+$ with $n \geq k$ is

$$P_n^k = \frac{n!}{(n-k)!}.$$

- $0!$ is defined to be equal to 1.

Definition 3.4.3. A **combination** is an unordered set. The number of combinations of size $k \in \mathbb{N}_+$ that can be formed from a set of size $n \in \mathbb{N}_+$ with $n \geq k$ is

$$C_n^k = \frac{n!}{k!(n-k)!}.$$

- By Definition 3.4.2 and Definition 3.4.3, for any $n, k \in \mathbb{N}_+$ with $n \geq k$,

$$C_n^k = \frac{P_n^k}{k!}.$$